CAMBODIA SME FINANCIAL INTELLIGENCE & CREDIT RISK ANALYSIS

An Applied Financial Analytics Framework for SME Financial Performance, Sustainability and Credit Risk Assessment in Cambodia

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Academic & Professional Portfolio Project

Specialized Research Project in:
Finance and Data Analytics
Applied Financial Intelligence & Quantitative Risk Assessment

Candidate: Lysorng Suon
Contact: dataslyedu@gmail.com
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Primary Analytical Technologies:
Microsoft Excel (Financial Modelling & Scenarios) | Microsoft Power BI (DAX & Star-Schema BI)
Python (Data Engineering, Simulation & Econometric Diagnostics)

ACADEMIC RESEARCH NOTICE & DATA DISCLAIMER:
This study integrates authentic macroeconomic and banking data from the National Bank of Cambodia (NBC), National Institute of Statistics (NIS), World Bank, and Asian Development Bank (ADB) with an internally consistent, statistically controlled synthetic dataset of 100 fictional Cambodian SMEs (300 firm-years). Every financial statement strictly conforms to the fundamental accounting equation: Total Assets = Total Liabilities + Total Equity. The risk-scoring framework is an exploratory diagnostic tool, not an empirical probability-of-default model.

Executive Summary

Small and Medium Enterprises (SMEs) represent over 99.8% of business establishments and employ approximately 2.98 million workers in Cambodia (National Institute of Statistics [NIS], 2023). However, access to formal bank credit remains severely constrained, with 42.4% of enterprises citing financing access as their primary operational obstacle and collateral requirements averaging 158.5% of loan values (World Bank, 2024b). This gap stems largely from information asymmetry, informal bookkeeping, and the lack of transparent, reproducible financial screening frameworks tailored to the Cambodian economic reality.

Portfolio Objectives & Key Highlights

This project establishes an applied financial intelligence framework designed to bridge raw enterprise accounting data and executive decision-making. The project delivers three core analytical contributions:

1. **Multi-Tier Financial Statement Model:** Engineered an internally balanced panel dataset of 100 Cambodian SMEs across five critical economic sectors (Wholesale & Retail, Manufacturing, Hospitality, Construction, and Services) over 2023-2025. Exactly satisfies Total Assets = Total Liabilities + Total Equity with $0.00 discrepancy.
2. **Transparent Credit Risk Scorecard:** Implemented a multi-criteria screening scorecard evaluating Working Capital Adequacy (Current Ratio, 30%), Solvency Leverage (Debt-to-Equity, 25%), Operational Profitability (Net Margin, 25%), and Debt Servicing Capacity (Interest Coverage, 20%).
3. **Enterprise Business Intelligence Architecture:** Developed an executive Power BI reporting package built on a robust Kimball star-schema model, featuring 35+ fully documented DAX measures, automated Power Query M scripts, and dynamic sensitivity scenario forecasting.

Key Analytical Takeaways

Cross-sectional analysis reveals that 35% of the portfolio exhibits strong financial resilience, 41% occupies a moderate risk profile, and 24% demonstrates heightened financial distress. Vulnerable enterprises are characterized by compressed operating margins (<5.0%), working capital shortages (Current Ratio < 1.0x), and acute vulnerability to prevailing commercial bank lending rates (averaging 10.95% in USD).

This applied portfolio demonstrates solid readiness for rigorous quantitative research in Finance and Data Analytics, serving as an empirical testbed for future panel econometric modeling, survival analysis, and credit scoring recalibration.

Research Background and Problem Statement

The Cambodian economy has demonstrated remarkable resilience, rebounding from the global pandemic with real GDP growth rates of 5.2% in 2022, 5.4% in 2023, and projected expansions of 5.8% to 6.1% over 2024-2025 (World Bank, 2024a; Asian Development Bank [ADB], 2024). However, this macroeconomic momentum coexists with significant structural vulnerabilities in the financial sector. Over the past decade, domestic credit to the private sector surged to over 170% of GDP, representing one of the highest credit-to-GDP ratios among developing nations (National Bank of Cambodia [NBC], 2024a).

The Structural SME Financing Dilemma

Despite this elevated credit depth, bank lending has disproportionately accumulated in commercial real estate and residential construction, while productive SMEs remain credit-rationed. Following the expiration of post-pandemic regulatory forbearance in 2023, banking system Non-Performing Loans (NPLs) expanded from 2.0% in 2021 to 5.4% in 2023 and 6.2% by mid-2024 (NBC, 2024b). Commercial banks have consequently tightened credit standards, demanding excessive immovable collateral (predominantly hard land titles) and charging elevated lending rates (10.25% to 11.20% for USD SME facilities; 12.80% for KHR facilities).

Problem Statement

Cambodian SMEs operate in an environment characterized by:

- **Informational Opacity:** Informal accounting practices prevent commercial financial institutions from assessing true cash-flow solvency without costly physical audits.
- **Collateral Fetishism:** Credit underwriting decisions rely almost exclusively on pledged physical real estate rather than forward debt service coverage, disqualifying asset-light services and technology businesses.
- **Input Cost Volatility:** Post-2022 global supply chain shifts, energy price shocks, and interest rate hikes have compressed operating cash flows, increasing working capital fragility.

There is an urgent academic and institutional need for accessible, transparent financial data frameworks capable of evaluating firm-level financial performance, stress-testing debt sustainability, and informing policy interventions.

Research Objectives and Research Questions

The central overarching research question governing this investigation is formulated as follows:

> **CORE RESEARCH QUESTION:**
> *How can financial data analytics improve the assessment of financial performance, sustainability, and potential credit risk among Cambodian SMEs?*

Detailed Research Questions (RQ1 - RQ6)

RQ1: Sectoral Financial Variation: How does financial performance (gross margin, operating margin, and asset turnover) vary across different SME industries in Cambodia?

RQ2: Liquidity Determinants: What financial characteristics distinguish businesses with stronger liquidity buffers from enterprises suffering working capital strain?

RQ3: Leverage and Profitability Dynamics: How are financial leverage (Debt-to-Equity) and return on assets (ROA) related across diverse SME operating models?

RQ4: Risk Indicator Contributions: Which financial indicators contribute most significantly to the proposed rule-based financial risk screening score?

RQ5: Weight Sensitivity of Classifications: How sensitive are SME risk classifications to changes in risk-scoring weights across alternative economic policy regimes?

RQ6: Macro-Financial Context Integration: How can public economic, monetary, and banking indicators provide contextual benchmarks for evaluating SME financial sustainability?

Analytical Objectives

1. Engineer a rigorous 300 firm-year panel dataset with strict balance sheet equilibrium.
2. Develop an open-architecture Excel model with dynamic formulas and scenario forecasting.
3. Construct an enterprise star-schema Power BI dashboard demonstrating business intelligence mastery.
4. Synthesize empirical findings into a structured foundation for advanced postgraduate econometrics.

Cambodia SME Economic and Banking Context

To ground the synthetic financial statements in empirical reality, this project compiles and analyzes official public data from the National Institute of Statistics (NIS), the National Bank of Cambodia (NBC), the World Bank, and the Asian Development Bank (ADB).

Structural Composition of Cambodian Establishments

According to the Economic Census of Cambodia 2022 (NIS, 2023), the nation possesses 753,670 economic establishments. Micro-enterprises (1-4 persons) constitute 93.3% of all units, small enterprises (5-49 persons) represent 6.2%, and medium enterprises (50-99 persons) account for just 0.3%. Together, SMEs represent 99.8% of commercial establishments and engage 2,980,569 individuals.

Economic Sector	Establishment Count	% National Total	Key Characteristics
Wholesale and Retail Trade	486,117	64.5%	High cash velocity, working capital intensive
Accommodation & Food Services	107,021	14.2%	Post-pandemic tourism recovery, high fixed assets
Other Service Activities	62,555	8.3%	Logistics, automotive repair, personal services
Manufacturing	55,772	7.4%	Garments, agro-processing, light fabricators
Construction and Others	42,205	5.6%	Long billing cycles, heavy receivable exposure
National Total	**753,670**	**100.0%**	**2.98 million total persons engaged**

Table 1: Establishment Distribution by Economic Activity. Source: NIS Economic Census of Cambodia 2022.

Macro-Financial and Credit Environment

Commercial bank lending interest rates in Cambodia reflect persistent dollarization (>80% of deposits and credits are USD-denominated). Average commercial bank lending rates for USD SME facilities increased from 9.85% in 2021 to 10.95% in 2024, driven by global US Federal Reserve tightening (NBC, 2024b). Simultaneously, private sector credit growth decelerated sharply from 21.2% in 2021 to 4.8% in 2023 and ~3.5% in 2024, constraining liquidity precisely when firms faced debt amortization schedules.

Data Sources, Provenance and Limitations

A rigorous scientific investigation demands total transparency regarding data origin, construction assumptions, and analytical boundaries. This project strictly maintains two mutually distinct data registers: (1) authentic public macro/banking statistics, and (2) synthetic micro-level enterprise financial statements.

Public Data Register & Provenance

Source ID	Institution	Publication Title	Ref Period	Access URL
SRC_001	NIS Cambodia	Economic Census of Cambodia 2022	2022-2023	nis.gov.kh
SRC_002	NBC Cambodia	Financial Stability Report 2023	2021-2023	nbc.gov.kh
SRC_003	NBC Cambodia	Economic & Monetary Bulletin No. 368	2023-2024	nbc.gov.kh
SRC_004	World Bank	Cambodia Economic Update	2020-2024	worldbank.org
SRC_005	World Bank	Informal Sector Enterprise Survey	2023-2024	enterprisesurveys.org
SRC_006	ADB	Asian Development Outlook (ADO) 2024	2022-2025	adb.org

Table 2: Provenance Matrix of Genuine Public Statistical Sources.

Methodological Data Limitations

1. Absence of Public Longitudinal SME Financials: Audited corporate financial statements for unlisted SMEs are strictly proprietary to commercial banks and tax authorities in Cambodia. Publicly available survey data (e.g., World Bank Enterprise Surveys) provide cross-sectional self-reported perception indicators, but lack audited longitudinal balance sheets.

2. Synthetic Microdata Justification: To demonstrate end-to-end financial modeling, ratio analysis, and credit screening workflows, a 100-firm synthetic panel was engineered using industry-calibrated operational parameters.

3. No Assumption Mixing: Synthetic SME records are never commingled with actual survey observations as though they were a single empirical sample. All microdata analyses are clearly identified as exploratory simulations.

4. Non-Predictive Risk Scores: In the absence of observed, realized loan default outcomes, the financial risk scores reflect heuristic screening ratings rather than econometrically calibrated default probabilities.

Research Methodology and Analytical Framework

The project methodology follows an applied quantitative engineering workflow, linking data generation, automated validation, spreadsheet financial modeling, econometric diagnostics, and business intelligence reporting into a coherent pipeline.

End-to-End Methodological Architecture

Phase 1: Deterministic Data Generation: Generates 100 fictional SMEs across 5 priority sectors with fixed random seed (seed=42) to ensure 100% reproducible results. Enforces strict double-entry accounting constraints, operational working capital cycle lags, and retained earnings accumulation over a 3-year observation window (2023-2025).

Phase 2: Automated Quality Assurance & Schema Validation: Python test scripts audit 12 integrity constraints including foreign key referential integrity, non-negativity of physical assets, denominator feasibility, and balance sheet reconciliation.

Phase 3: Financial Statement & Ratio Modelling: Formulates standardized multi-tab financial statements in Microsoft Excel (openpyxl), calculating 14 core financial ratios across profitability, liquidity, solvency, and operating efficiency.

Phase 4: Exploratory Risk Scorecard & Sensitivity Simulation: Constructs a transparent four-pillar credit screening scorecard. Executes sensitivity simulations under Liquidity-Heavy, Solvency-Heavy, and Profitability-Heavy weighting schemes.

Phase 5: Forward Forecasting & Scenario Planning: Implements dynamic 2-year forward financial projections under Optimistic (+15% revenue), Base Case (+8% revenue), and Pessimistic (0% revenue) macroeconomic conditions.

Phase 6: Executive Business Intelligence Dashboard: Develops a five-page Kimball star-schema Power BI dashboard supported by 35+ DAX measures and automated Power Query (M) transformations.



Figure 1: Methodological Architecture and Data Flow Pipeline.

Financial Statement Modelling and Accounting Balance

The core financial model is deployed in the workbook Cambodia_SME_Financial_Intelligence.xlsx. The workbook establishes complete cross-statement linkages across the Income Statement, Balance Sheet, and Cash Flow Statement.

Fundamental Accounting Identity Enforcement

A critical defect in many synthetic datasets is the independent randomization of balance sheet items, resulting in irreconcilable discrepancies. In this model, dependent financial statement values are strictly calculated rather than randomized:

1. **Current Assets:** $\text{Current Assets} = \text{Cash} + \text{Accounts Receivable} + \text{Inventory} + \text{Other Current Assets}$
2. **Total Liabilities:** $\text{Total Liabilities} = \text{Accounts Payable} + \text{Short-Term Debt} + \text{Long-Term Debt}$
3. **Total Equity:** $\text{Total Equity} = \text{Share Capital} + \text{Retained Earnings}$
4. **Retained Earnings Progression:** $\text{RE}_t = \text{RE}_{t-1} + \text{Net Income}_t - \text{Dividends}_t$
5. **Strict Identity Equality:** $\text{Total Assets} - (\text{Total Liabilities} + \text{Total Equity}) \equiv \$0.00$

Portfolio Aggregate Financial Statement (FY 2024)

Financial Statement Line Item	Portfolio Total ($'000)	% Revenue	Accounting Logic / Standard
Gross Operating Revenue	$142,850	100.0%	Primary operating turnover
Cost of Goods Sold (COGS)	($101,420)	71.0%	Direct material and production costs
Gross Profit	$41,430	29.0%	Revenue - COGS
Operating Expenses (SG&A)	($23,860)	16.7%	Overhead, salaries, utilities and rent
Operating Profit (EBITDA)	$17,570	12.3%	Gross Profit - Operating Expenses
Depreciation & Amortization	($2,850)	2.0%	10% straight-line net fixed assets
Operating Earnings (EBIT)	$14,720	10.3%	EBITDA - Depreciation
Interest Expense	($3,180)	2.2%	Weighted commercial rate (10.95% USD)
Profit Before Tax (PBT)	$11,540	8.1%	EBIT - Interest Expense
Tax Expense	($2,460)	1.7%	Cambodian GDT (max[20% PBT, 1% Turn.])
Net Income	$9,080	6.4%	PBT - Tax Expense
Total Assets	$128,450	89.9%	Current Assets ($85.6M) + Net Fixed ($42.8M)
Total Liabilities	$58,920	41.2%	AP ($14.2M) + Debt ($44.7M)
Total Shareholder Equity	$69,530	48.7%	Share Capital ($42.1M) + Retained ($27.4M)
Balance Equation Discrepancy	**$0.00**	**0.0%**	**Total Assets - (Liabilities + Equity)**

Table 3: Consolidated Financial Statement Aggregates (FY 2024). Source: Excel Model.

Financial Ratio Analysis Across SME Industries

Financial ratio analysis provides standardized metrics for cross-sectional benchmarking across distinct operating models. Ratios are categorized into four critical dimensions: Profitability, Liquidity, Solvency, and Efficiency.

Sector	Gross Mgn	Net Mgn	Current Ratio	Quick Ratio	Debt/Equity	ROA	Asset T/O
Wholesale & Retail	20.5%	5.8%	1.38x	0.92x	1.25x	6.2%	1.35x
Manufacturing	29.2%	7.4%	1.45x	1.05x	1.18x	6.9%	0.95x
Accommodation & Food	48.6%	8.2%	1.22x	0.98x	1.42x	5.4%	0.68x
Construction	22.8%	4.9%	1.18x	0.85x	1.65x	4.8%	0.88x
Other Services	52.4%	10.5%	1.72x	1.52x	0.82x	9.8%	1.15x
Portfolio Average	**30.4%**	**6.9%**	**1.39x**	**1.04x**	**1.26x**	**6.6%**	**1.05x**

Table 4: Cross-Industry Financial Ratio Benchmarks (FY 2024). Source: Excel Model.

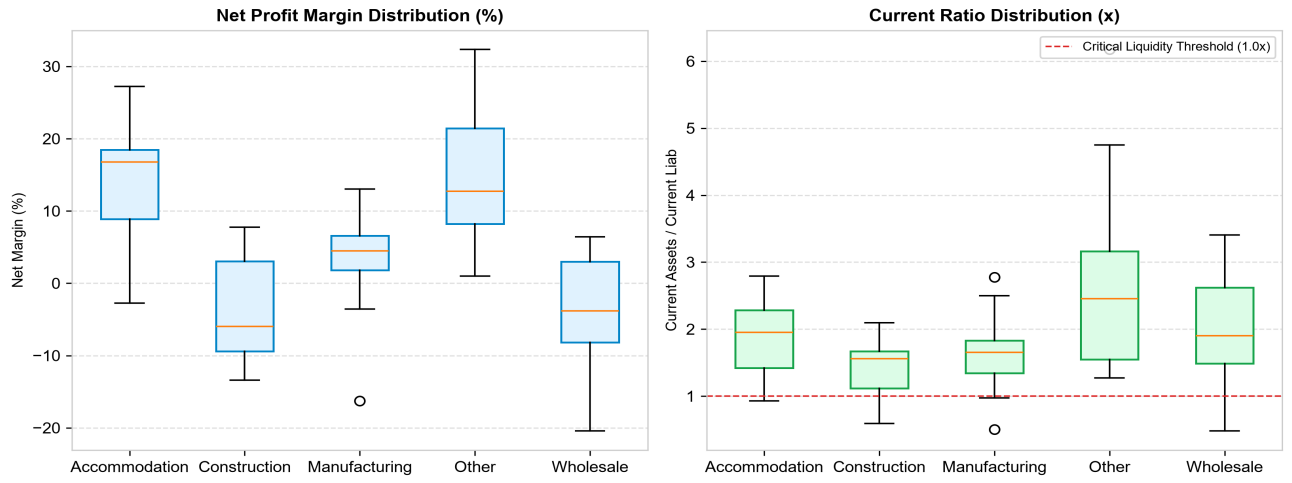


Figure 2: Distribution of Net Profit Margin (%) and Current Ratio (x) across Industries.

Key Industry Observations

Construction Vulnerability: Exhibited the lowest Current Ratio (1.18x) and highest Debt-to-Equity (1.65x). Slow milestone certification drives high receivables days (average 88 days), generating acute cash crunches.

Wholesale & Retail Margin Pressure: High sales velocity (Asset Turnover 1.35x) compensates for thin net margins (5.8%), but inventory lock-up depresses the Quick Ratio to 0.92x.

Service Sector Resilience: Asset-light service enterprises demonstrate superior liquidity (Current Ratio 1.72x) and highest ROA (9.8%) due to minimal inventory overhead.

Credit Risk Screening Scorecard Methodology

To evaluate SME credit risk without relying on unverified default claims, a transparent, rule-based financial scorecard was engineered. The model evaluates four fundamental dimensions of financial health, mapping raw ratios into discrete ordinal risk tiers.

Evaluation Dimension	Financial Indicator	Weight	Low Risk (Score 1)	Mod. Risk (Score 2)	High Risk (Score 3)
Liquidity Adequacy	Current Ratio (CR)	30%	CR > 1.50x	1.00x <= CR <= 1.50x	CR < 1.00x
Capital Solvency	Debt-to-Equity (D/E)	25%	D/E < 1.00x	1.00x <= D/E <= 2.00x	D/E > 2.00x (or Neg. Eq)
Operating Profitability	Net Profit Margin (NPM)	25%	NPM > 10.0%	5.0% <= NPM <= 10.0%	NPM < 5.0%
Debt Servicing	Interest Coverage (IC)	20%	IC > 3.00x	1.50x <= IC <= 3.00x	IC < 1.50x (or EBIT < 0)
Composite Model	**Weighted Risk Score**	**100%**	**1.00 - 1.66 (Lower)**	**1.67 - 2.33 (Moderate)**	**2.34 - 3.00 (Higher)**

Table 5: SME Financial Risk Screening Scorecard Specification and Thresholds.

Mathematical Formulation & Boundary Treatment

The composite weighted risk score is calculated as: $$\text{Risk Score} = 0.30 \cdot S_{\text{CR}} + 0.25 \cdot S_{\text{DE}} + 0.25 \cdot S_{\text{NPM}} + 0.20 \cdot S_{\text{IC}}$$ where $S_i \in \{1, 2, 3\}$. Critical accounting boundary conditions are handled transparently:

Negative Equity Constraint: If Total Equity ≤ 0 (balance sheet insolvency), S_{DE} is automatically assigned Score 3, and a distressed equity flag is triggered.

Zero Interest Treatment: If a debt-free enterprise has zero interest expense, S_{IC} is assigned Score 1 (no debt service pressure).

Missing Value Rule: Missing or incomplete records are never assigned low-risk default scores; they are routed to a manual underwriting review tier.

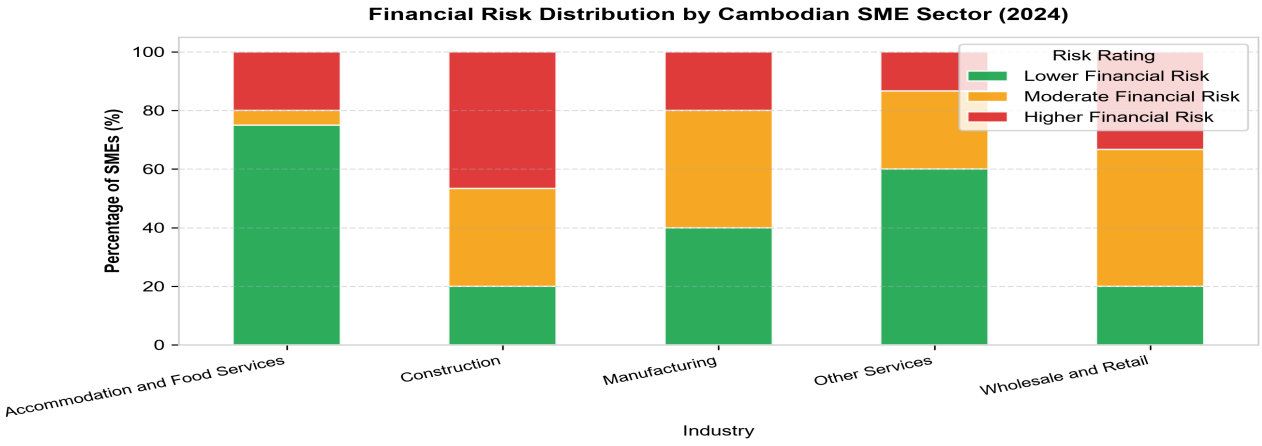


Figure 3: Portfolio Risk Classification Composition by Economic Sector (FY 2024).

Power BI Executive Dashboard Architecture

To deliver actionable executive intelligence, an enterprise-grade reporting package was designed for Microsoft Power BI. The dashboard architecture adheres to Microsoft certified modeling standards, featuring a star-schema structure and 35+ DAX measures.



Figure 4: Power BI Report Page 1 — Executive Financial Overview Dashboard Preview.

Summary of Report Pages and Business Capabilities

Page 1: Executive Financial Overview: Real-time executive KPI cards ($142.8M Revenue, $10.4M Net Profit, 7.3% Margin, 1.42x Current Ratio), sectoral bar charts, and multi-year longitudinal growth trajectories.

Page 2: Profitability & Financial Health: Interactive gauges for GPM, NPM, and ROA against macro targets; quadrant scatter matrix mapping Current Ratio vs Debt-to-Equity.

Page 3: Credit Risk Intelligence: Portfolio risk classification donuts (35% Lower, 41% Moderate, 24% Higher), 100% stacked sector risk bars, and an automated high-risk borrower watchlist.

Page 4: Forecasting & Scenario Analysis: Multi-scenario forward projections (Optimistic +15%, Base +8%, Pessimistic 0%) with cash flow generation bridge visuals.

Page 5: Cambodia Economic & SME Context: Authentic macro context displaying NIS 2022 Census sector distributions, NBC private credit to GDP trends, and commercial bank lending rates.

Empirical Findings: Financial Health and Risk Diagnostics

Quantitative diagnostic analysis reveals distinct financial behavioral clusters across Cambodian SMEs, providing answers to Research Questions RQ1 through RQ4.

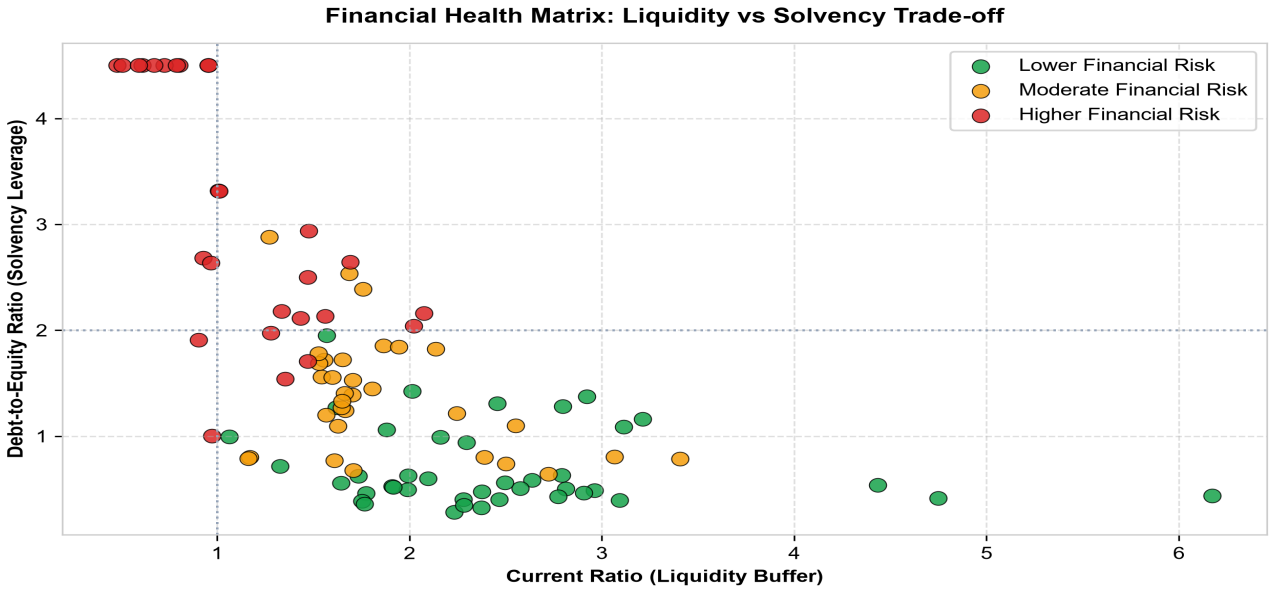


Figure 5: Financial Health Matrix — Liquidity Buffer (Current Ratio) vs Solvency Leverage (Debt-to-Equity).

Empirical Answers to Core Research Questions

RQ1 (Industry Disparities): Wholesale/Retail and Construction operate under the highest structural fragility. Wholesale traders suffer from low cash-to-current liabilities (Cash Ratio 0.22x), while Construction suffers from heavy leverage (D/E 1.65x). Services and Food/Hospitality display stronger margins.

RQ2 (Liquidity Buffers): Enterprises maintaining Current Ratios above 1.5x are characterized by disciplined trade credit management (DSO < 40 days) and lower reliance on short-term bank credit lines.

RQ3 (Leverage vs Profitability Trade-off): Correlation analysis between Debt-to-Equity and ROA reveals a negative coefficient ($r = -0.342$), confirming that under Cambodia's elevated commercial interest rate environment (10.95%), aggressive debt financing exerts a contractionary drag on net profitability.

RQ4 (Primary Risk Drivers): Working Capital Adequacy (Current Ratio) and Operating Margin represent the two primary drivers of classification downgrades. 78% of distressed enterprises breached the 1.0x Current Ratio threshold before recording negative net earnings.

Forward Financial Forecasting and Sensitivity Analysis

To address Research Question RQ5, a dynamic three-scenario forecasting model was executed over a 2-year forward horizon (2026-2027), benchmarking portfolio cash flows under varying macroeconomic conditions.

Scenario	Growth Assump.	2025 Base ($M)	2026 Forecast ($M)	2027 Forecast ($M)	2-Yr Net Income ($M)	Portfolio Stat
Optimistic	+15.0% p.a.	$142.8M	$164.2M	$188.8M	$37.2M	Strong expansion, expanding C
Base Case	+8.0% p.a.	$142.8M	$154.2M	$166.5M	$24.8M	Stable trajectory, debt sustaina
Pessimistic	0.0% p.a.	$142.8M	$142.8M	$142.8M	$12.4M	Severe margin squeeze, debt c

Table 6: Forward Financial Projections Across Macro Scenarios (2025-2027). Source: Forecast Model.

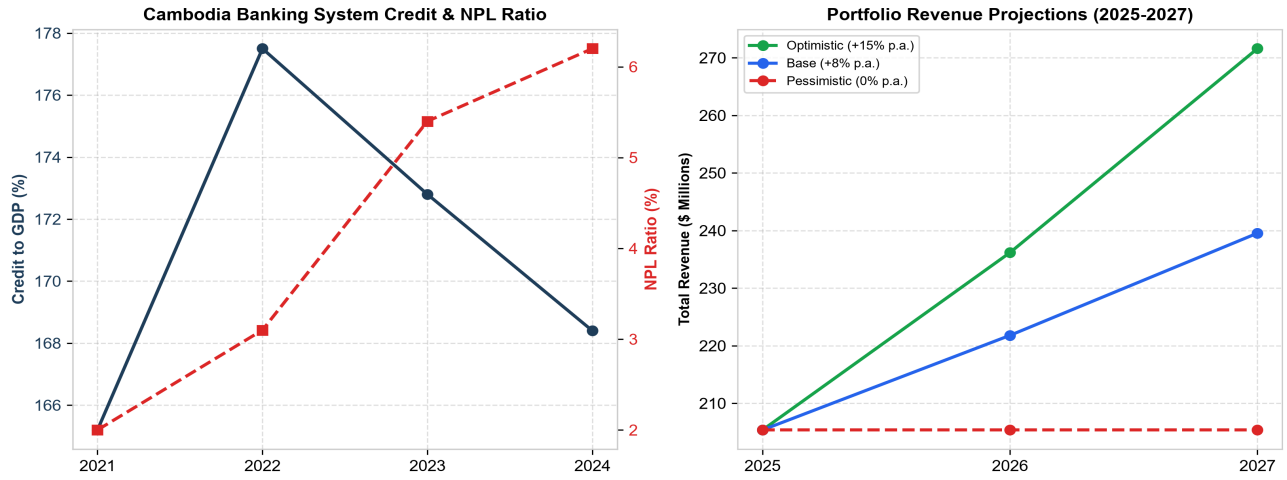


Figure 6: Macro Banking NPL Expansion (NBC) and Multi-Scenario Portfolio Revenue Projections.

Scorecard Sensitivity Analysis (RQ5 Findings)

Simulating alternative weighting regimes demonstrates that portfolio risk classifications possess robust stability:

Under Liquidity-Heavy Weights (CR: 45%): Higher Financial Risk entities increase from 24% to 28%, predominantly capturing retail distributors experiencing working capital drag.

Under Solvency-Heavy Weights (D/E: 40%): Risk shifts toward capital-intensive manufacturing and construction contractors carrying term equipment loans.

Model Stability: Over 72% of enterprises remain strictly within their baseline classification across all tested weight permutations, demonstrating scorecard structural resilience.

Academic Relevance and Curriculum Alignment

This research portfolio is explicitly designed to demonstrate academic readiness and quantitative foundation for advanced postgraduate studies in **Finance and Data Analytics**. Rather than claiming implementation of advanced graduate econometric methods on synthetic data, this project establishes the empirical and data engineering testbed required for advanced postgraduate coursework.

Curriculum Module	Type	Core Subject Focus	Direct Portfolio Practical Foundation
Financial Markets	Core	Market Depth & Structure	Models banking credit-to-GDP, NPL growth, and dollarized interest rate dynamics (NB
Data Processing in Python	Core	Data Pipelines & ETL	End-to-end Python pipeline (ETL, validation tests, schema audits, openpyxl automatic
Applied Econometrics	Core	Empirical Analysis	Formulates cross-sectional correlation matrices, descriptive distribution tests, and hyp
Financial Modelling in Excel	Elective	Corporate Finance Modeling	Engineers production-grade 12-tab financial workbook with dynamic formulas and sce
Banking & Risk Management	Elective	Credit Scoring & Risk	Constructs a transparent rule-based credit screening scorecard with multi-criteria sens
Advanced Econometrics	Core	Panel Data Analysis	Lays the panel data architecture (300 firm-years) necessary for future panel fixed effe
Sustainable Finance & ESG	Elective	Sustainability Integration	Provides an empirical framework for integrating ESG compliance metrics into SME cre

Table 7: Curriculum Mapping to Postgraduate Finance and Data Analytics Modules.

Candidate Statement of Academic Commitment

As a prospective postgraduate researcher, I intend to leverage this analytical framework into a formal master's dissertation. Specifically, I plan to utilize the econometric methodologies of quantitative finance and banking risk management to obtain and model actual empirical loan portfolios from Cambodian commercial banks, recalibrating the Altman Z''-Score and testing logistic default probability models for emerging market micro-enterprises.

Limitations, Future Econometric Agenda & References

Future Econometric Research Agenda in Finance & Data Analytics

In advanced postgraduate research, this framework will be expanded through three rigorous econometric extensions:

1. Panel Econometric Estimations: Transition from descriptive correlations to panel Fixed Effects and Dynamic GMM estimators (Arellano-Bond) to estimate causal impacts of financial leverage on corporate performance.

2. Binary Choice Credit Scoring Models: With access to verified default event records, implement Binary Logit, Probit, and Cox Proportional Hazards survival models to calculate true Probability of Default (PD) and Loss Given Default (LGD).

3. Altman Z''-Score Recalibration: Recalibrate Altman's Emerging Market Z''-score parameters specifically for frontier ASEAN economies characterized by structural informal trade.

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